

System Architecture & Technology Stack

ITNEXT Global Careers — a modular-monolith career & global-mobility platform:
Next.js front end, FastAPI back end, load-balanced across replicas, containerised
for local development and Azure Container Apps.

ROUTE

LOCALHOST →
AZURE

CLASS

Modular
Monolith

GATE

nginx :8000

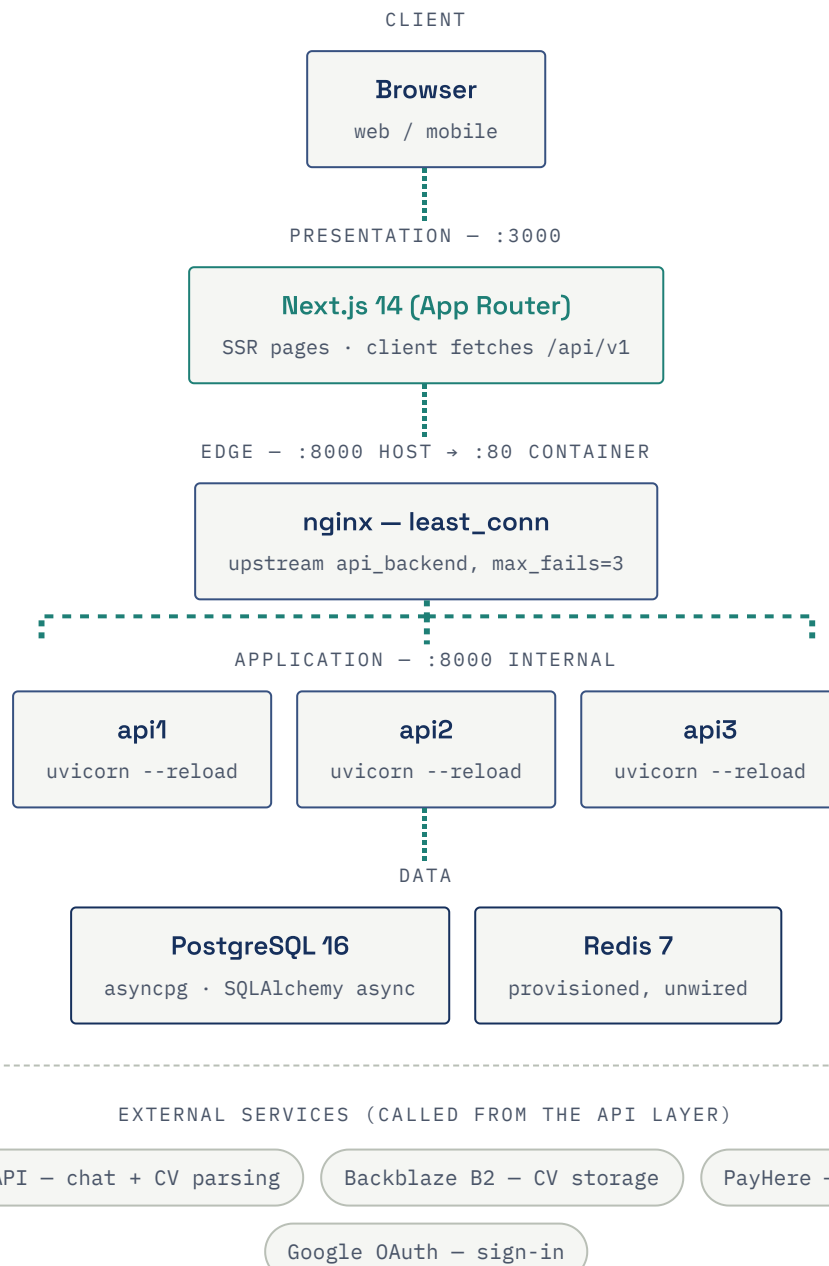
SEATS

3× API replica

01 Summary

The system is a career-services and global-mobility platform: users pick a service package, upload a CV, complete a profile questionnaire, and talk to an AI career assistant. It ships as a **Next.js** web app talking to a **FastAPI** API, both containerised. Locally, three identical API replicas sit behind an **nginx** load balancer so the request-spreading behaviour matches production. Data lives in **PostgreSQL**; a **Redis** container is provisioned in the stack but is not yet called from any application code. In production the containers deploy to **Azure Container Apps**, autoscaling between 1 and 5 replicas.

02 Request Path & Architecture



The frontend and the load balancer are reached on separate host ports (3000 and 8000) — the browser, not nginx, is what stitches them into one product.

03 Component Manifest

SERVICE	IMAGE / BUILD	ROLE	PORT (HOST→CTR)
db	postgres:16-alpine	Primary datastore	5433→5432
redis	redis:7-alpine	Provisioned cache/session store — no app code calls it yet	6379→6379
api1 / api2 / api3	./backend (python:3.12-slim)	FastAPI replica, identical, behind the LB	— (internal :8000)
nginx	nginx:alpine	Load balancer, least_conn across the 3 replicas	8000→80
web	./frontend (node:20-slim)	Next.js frontend, standalone server	3000→3000

The db port was remapped from the compose default 5432 to 5433 on this machine to avoid colliding with a native Windows PostgreSQL 18 service also bound to 5432.

04 Technology Stack

Frontend

Next.js	14.2.15	App Router, SSR + client components
React	18.3.1	UI runtime
TypeScript	5.6.3	Static typing (dev)
Tailwind CSS	3.4.14	Styling (dev)

Backend

Python	3.12	Runtime (slim container)
FastAPI	0.115.5	Web framework, modular-monolith routers
Uvicorn	0.32.1	ASGI server
Pydantic / pydantic-settings	2.12.5 / 2.14.2	Schemas & env-driven config
httpx	0.28.1	Outbound HTTP client

Data & persistence

PostgreSQL	16-alpine	Primary datastore
SQLAlchemy (asyncio)	2.0.36	ORM, async engine
asyncpg	0.30.0	Postgres driver
Alembic	1.14.0	Schema migrations — 6 revisions committed

Auth & security

python-jose	3.3.0	JWT issuance/verification, HS256, 24h expiry
passlib + bcrypt	1.7.4 / 4.3.0	Password hashing
Google OAuth	client-side	Optional "Continue with Google" — hidden unless a client ID is set

AI, storage & payments (external services)

Gemini API	google-genai 2.12.1	Career-assistant chat & CV parsing — model gemini-flash-lite-latest
Backblaze B2	boto3 1.35.68 (S3-compatible)	CV storage; falls back to local disk if unset
PayHere	sandbox	Package checkout

Infrastructure & delivery

Docker Compose	local	db, redis, 3× api, nginx, web
Azure Container Apps	production	Min 1 / max 5 autoscaled replicas
Azure Container Registry	itnextacr	Image storage for deploys
GitHub Actions	CI + CD	CI: ruff + pytest + npm build on push/PR. CD: make ship on a v* tag

05 Backend Module Map

The API is one deployable service, internally split into self-contained modules — each owns its models, schemas, service layer, and router, so any one could be extracted into its own service later without rewiring the rest.

auth

Signup / login, JWT issuance, Google OAuth

packages

Service-tier catalogue and selection

cv

Presigned upload, storage (B2/local), background parse

chat

AI career assistant (Gemini-backed, polling not WebSocket)

admin

Admin-role operations (no models of its own)

assessment

Professional-profile questionnaire, stored as JSONB

payments

Order creation, PayHere checkout

06 Known Gaps & Stubs

The system boots with zero external credentials by design — each of the following degrades to a stub rather than failing:

- 01 **Redis** is running in the stack but no module currently imports a Redis client — it is provisioned, not wired.
- 02 **CV parsing** is a marked TODO in `cv/service.py` — the Gemini call that turns an uploaded CV into structured JSON is not yet implemented.
- 03 **Chat** replies with a stub message until `GEMINI_API_KEY` is set — note the README's "Claude-powered" framing is aspirational; the wired implementation calls Gemini, not Anthropic.
- 04 **PayHere webhook** does not yet verify the `md5sig` signature before marking an order paid.
- 05 **Azure Blob Storage** has a config surface (`AZURE_STORAGE_CONNECTION_STRING`) but no implementation — only the B2 and local-disk backends are coded.

Compiled from repository inspection — `docker-compose.yml`, `backend/requirements.txt`, `frontend/package.json`, `backend/app` source, `.github/workflows` — 2026-07-18. Not a substitute for reading the code.